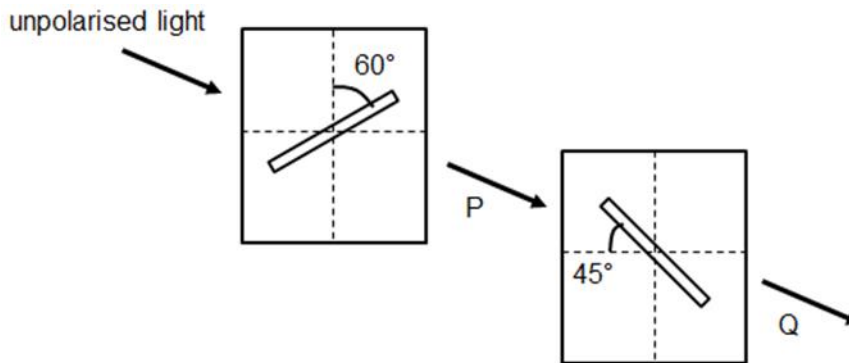


17

A narrow, parallel beam of unpolarised light is passed through two optical polarisers. The first polariser's transmission axis is oriented at 60° to the vertical, while the second polariser's transmission axis is oriented at 45° to the horizontal. The light at P has amplitude A .



What is the amplitude of the light at Q?

A

$A \cos 15^\circ$

B

$A \cos 45^\circ$

C

$A \cos 60^\circ$

D

$A \cos 75^\circ$

L2

Answer: D

Angle between the polarized light at P and transmission axis of the 2nd polarizer
 $= 180 - (60 + 45)$
 $= 75^\circ$

Amplitude of the light after passing through Q
 $= A \cos 75^\circ$

18

A fabric consists of closely spaced horizontal and vertical threads as shown